

PRESS METHODOLOGY MODERNIZATION

April 2025



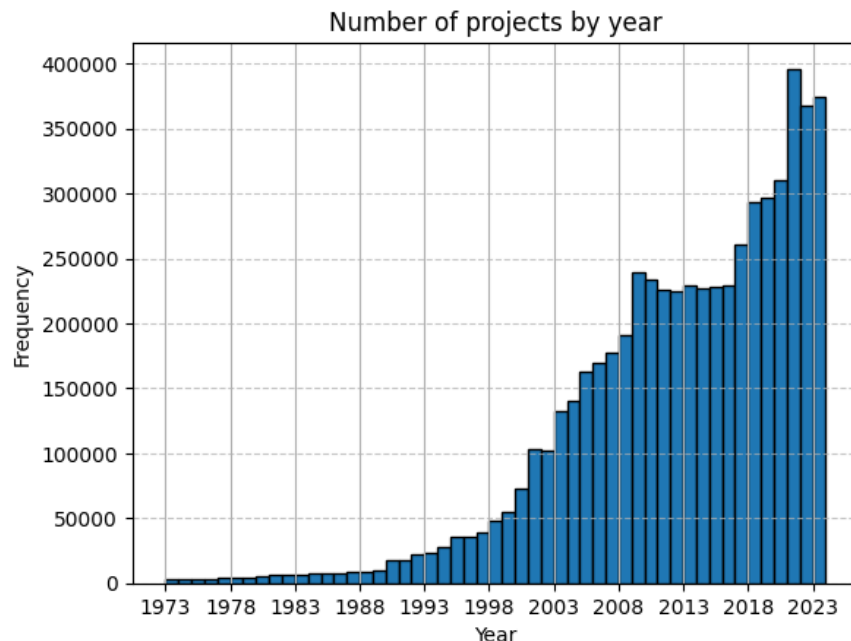
1. CRS Data
2. Methodological Requirements
3. Current Methodology
4. Methodology Proposal
 1. Classification method
 2. Summary new methodology
 3. Comparison current vs proposed methodology
5. Up for discussion

1. CRS Data

CRS data

entire CRS dataset:

- 5.8m observations (projects) for 98 variables from 1973-2023
- variables necessary for our classification methodology:
 - texts:** *project_title, short_description, long_description*
 - markers:** *purpose_code, channel_code, gender, rmnch*



available variables

```
[ 'agency_code', 'agency_name', 'aid_t', 'bi_multi', 'biodiversity', 'blended_finance_type', 'capital_expend', 'category', 'channel_code', 'channel_name', 'channel_reported_name', 'climate_adaptation', 'climate_mitigation', 'commitment_date', 'commitment_national', 'completion_date', 'crs_id', 'currency_code', 'de_donorcode', 'de_incomegroup_code', 'de_recipientcode', 'de_regioncode', 'desertification', 'dig', 'disability', 'disbursement_national', 'donor_code', 'donor_name', 'drr', 'environment', 'expected_start_date', 'finance_t', 'flow_code', 'flow_name', 'ftc', 'gender', 'geography', 'grant_equiv', 'incomegroup_code', 'incomegroup_name', 'initial_report', 'interest1', 'interest2', 'investment_project', 'keywords', 'ld_cflag', 'ld_cflag_name', 'long_description', 'number_repayment', 'nutrition', 'parent_channel_code', 'pba', 'project_number', 'project_title', 'ps_iflag', 'ps_iflag_name', 'psi_add_desc', 'psi_add_dev_obj', 'psi_add_type', 'purpose_code', 'purpose_name', 'recipient_code', 'recipient_name', 'region_code', 'region_name', 'repaydate1', 'repaydate2', 'rmnch', 'sd_gfocus', 'sector_code', 'sector_name', 'short_description', 'trade', 'type_repayment', 'usd_adjustment', 'usd_adjustment_defl', 'usd_amount_partial_tied', 'usd_amount_partial_tied_defl', 'usd_amount_tied', 'usd_amount_untied', 'usd_amount_untied_defl', 'usd_amounttied_defl', 'usd_arrears_interest', 'usd_arrears_principal', 'usd_commitment', 'usd_commitment_defl', 'usd_disbursement', 'usd_disbursement_defl', 'usd_expert_commitment', 'usd_expert_extended', 'usd_export_credit', 'usd_grant_equiv', 'usd_interest', 'usd_irtc', 'usd_outstanding', 'usd_received', 'usd_received_defl', 'year' ]
```

1. CRS Data

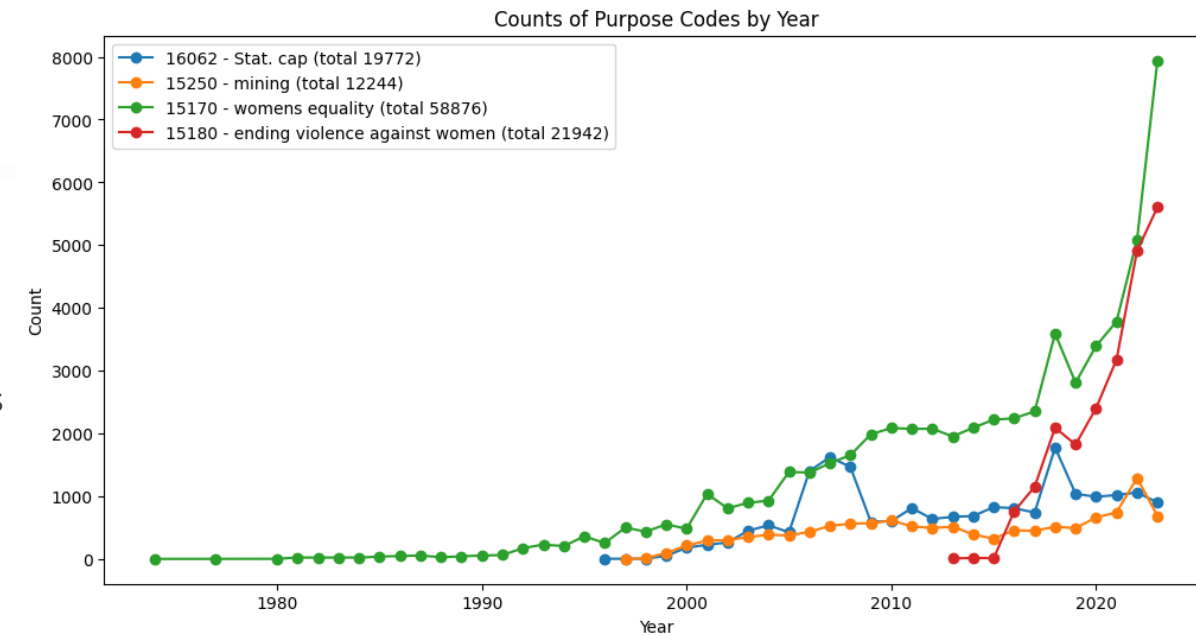
Purpose codes and other project marker

statistics marker

- purpose codes: 16062 – Statistical capacity building
15250 – Removal of land mines and explosive remnants of war (for blacklisting)

gender markers

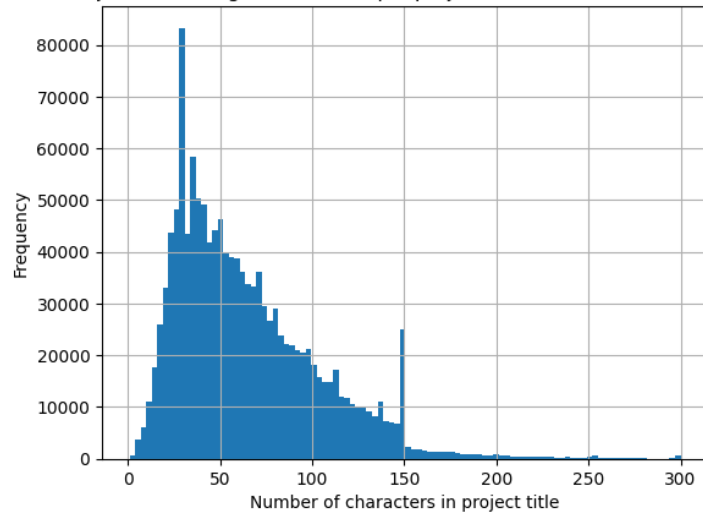
- purpose codes: 15170 - Women's equality organizations and institutions
15180 - Ending violence against women and girls
- Channel code: 41146 - UN woman
- CRS gender marker: 0: When gender equality is not targeted
1: When gender equality is a significant objective
2: When gender equality is a principal objective.
- policy marker: rmnch - reproductive, maternal, newborn and child health (since 2013)



1. CRS Data

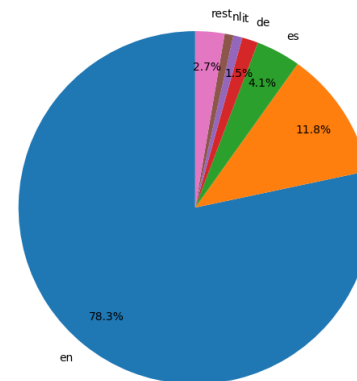
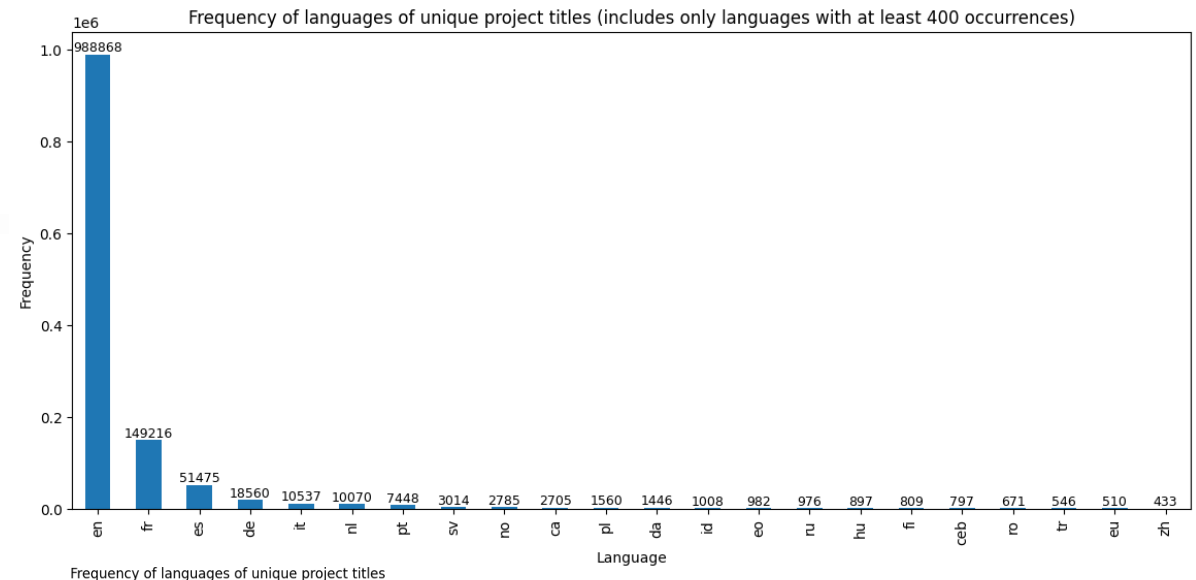
Project titles

Histogram of Project Title Lengths (total unique project titles: 1320652, None values: 681483)



Language analysis for project titles

using library **fasttext** (see [here](#) for comparison of popular language detection packages), all algorithms a bit shaky on very short texts



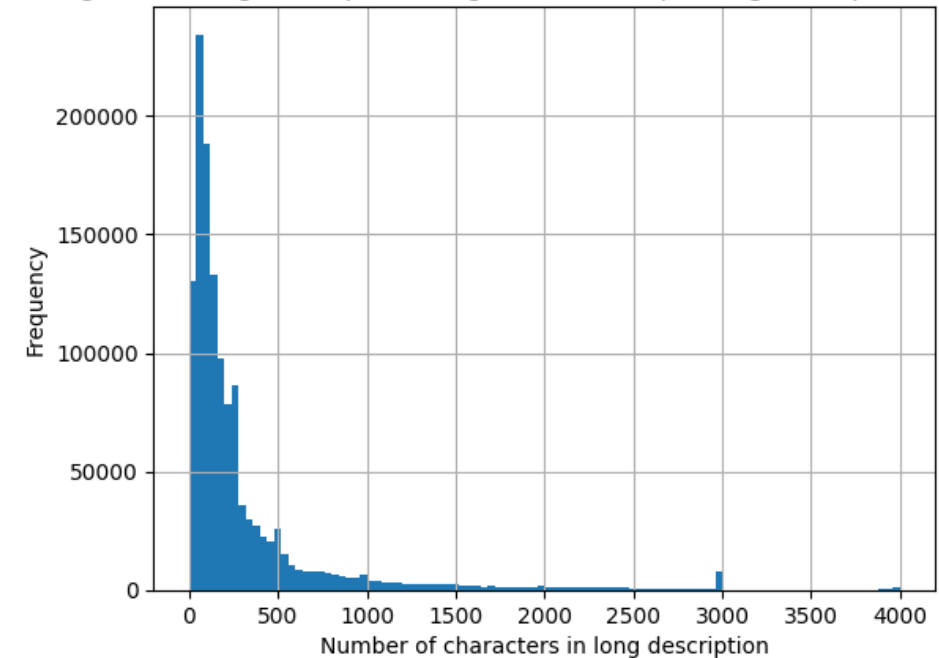
takeaway from language analysis: almost all project titles in either en/fr/es/de/it, rest misclassification due to too short project title or abbreviations used (e.g. “SEM. EVALUA. DIAKONIA SA”, “SUA&TARP II”)

Long descriptions

examples:

- *to strengthen the capacity of National Malaria Control Programs to generate, analyze, and utilize genomic data to support malaria control and elimination efforts in 16 countries in sub-saharan Africa*
- *to improve donor funding alignment and coordination of global levers in support of health sector goals*
- *Energy efficiency investment (steam plant). Steam is used to produce energy. The current 20MW steam boiler and associated facility was built in the 1960's and is operated on a petroleum basis. This investment will finance and install a new steam plant powered by natural gas. This reduces pollutant emissions. Environmental effects: The new steam boiler system improves power generation through improved power generation technology while increasing energy efficiency. The project thus supports the fight against climate change. SDGs 1, 8 and 17 apply.*
- *CENTRALE GEOTHERMIQUE DE LAHENDONG*
- *Development of educated classes in society, promotion of international exchange and networking*
- *Fernão Mendes Pinto scholarship*
- *to provide direct grant to Borno State for PHC strengthening leveraging lessons from the RI basket fund model and the NSHIP implementation in the state*

Histogram of Long Description Lengths (total unique long descriptions: 1289142)



2. Methodological Requirements

Due to the structure of our data with titles, short and long descriptions in multiple language, as well as projects marked as statistical projects through purpose code 16062, two types of methodological requirements can be distinguished:

Requirements due to data structure:

1. Multilingual Reporting

classification methodology should work on multiple languages

2. Positive-unlabeled learning problem

in our case, we have a robust set of labeled projects for which a reporter has assigned a positive label (for statistics: purpose code 16062), but no negatively labeled projects for which we definitively know they have no statistics/data component

Requirements from us:

3. Accuracy

to us, accuracy is more important than e.g. computing time

4. Interpretability

classification methodology should offer the possibility to provide insight into how it arrives at a classification (e.g. relative importance of words for XGBoost, i.e. which words are weighted how much)

5. Variance estimation

it would be helpful to quantify the uncertainty for the numbers we report (e.g. 875m $\pm$ 0.2m)



All steps

Preparation

1. **Downloading CRS datasets:** 1 year, 1 dataset
2. **Create project ID:** hashed string based on projectnumber, crsid, year, purposecode, usd_disbursement, recipientcode (df_crs_hash)

A – Title pattern matching ([see methodology note for more details](#))

1. **Clean text:** remove punctuation & numbers, lowercase letters in titles, short description, long description
2. **Language detection of titles:** uses CLD2 to detect the language of the project title
3. **Detecting keywords in titles:** keyword matching for statistics & gender from keyword lists (see next slide)
4. **Marking projects with purpose code:** 16062 – Statistical capacity building, 15250 - Removal of land mines, 15170-15180 – Gender purpose codes
5. **Final marker:** (statistics keyword match **or** stat purpose code) & (**not** mining title & **not** mining purpose code) (similar for gender)

B – Text mining

idea: more projects have statistics component beyond A -> identification through long description

1. **Identify long descriptions for text mining:** use long description only if it's different from title and short description
2. **Calculation of odds ratio for every term in statistical project's long description compared to non-statistical project (no stat marker)**
3. **Using odds ratio to filter projects without statistics purpose code and statistics title**

3. Current Methodology

Keywords

statistics

- 4 languages (en, fr, es, de)
- 1 list for keywords,
1 list for acronyms,
1 blacklist (mining projects to be excluded, frequently include “survey”)

gender

- 4 languages (en, fr, es, de)
- 1 list for keywords,
1 list for acronyms,
no blacklist

keywords stat

statistics
survey
census
database
big data
data for decisions
civil registration
civil register
land registration
land register
cadaster
sdg indicator
information system
birth registration
birth register
business register
national account
price index
production index
data science
data for development
data journalism
data for education
education data
data for health
peacebuilding data
global data
global pulse

acronyms stat

dhs
crvs
cpi
d4d
paris21
gpsdd
odw
data4sdg
devinfo
nsds
afristat
ckan
lfs questionnaire
d4d
data4development
50X2030
50 by 2023
lfs
ophi

blacklist stat

land mine
small arm
demining
demine
landmine
explosive

keywords gender

women
female
girl
pregnant
marriage
birth
gender
sex
social protection
data disaggregation
Gender statistics
Gender data
Gender equality
feminicide
time use data
time use survey
unpaid work
care work
reproductive
intersectional
maternal
unpaid care
genital

acronyms gender

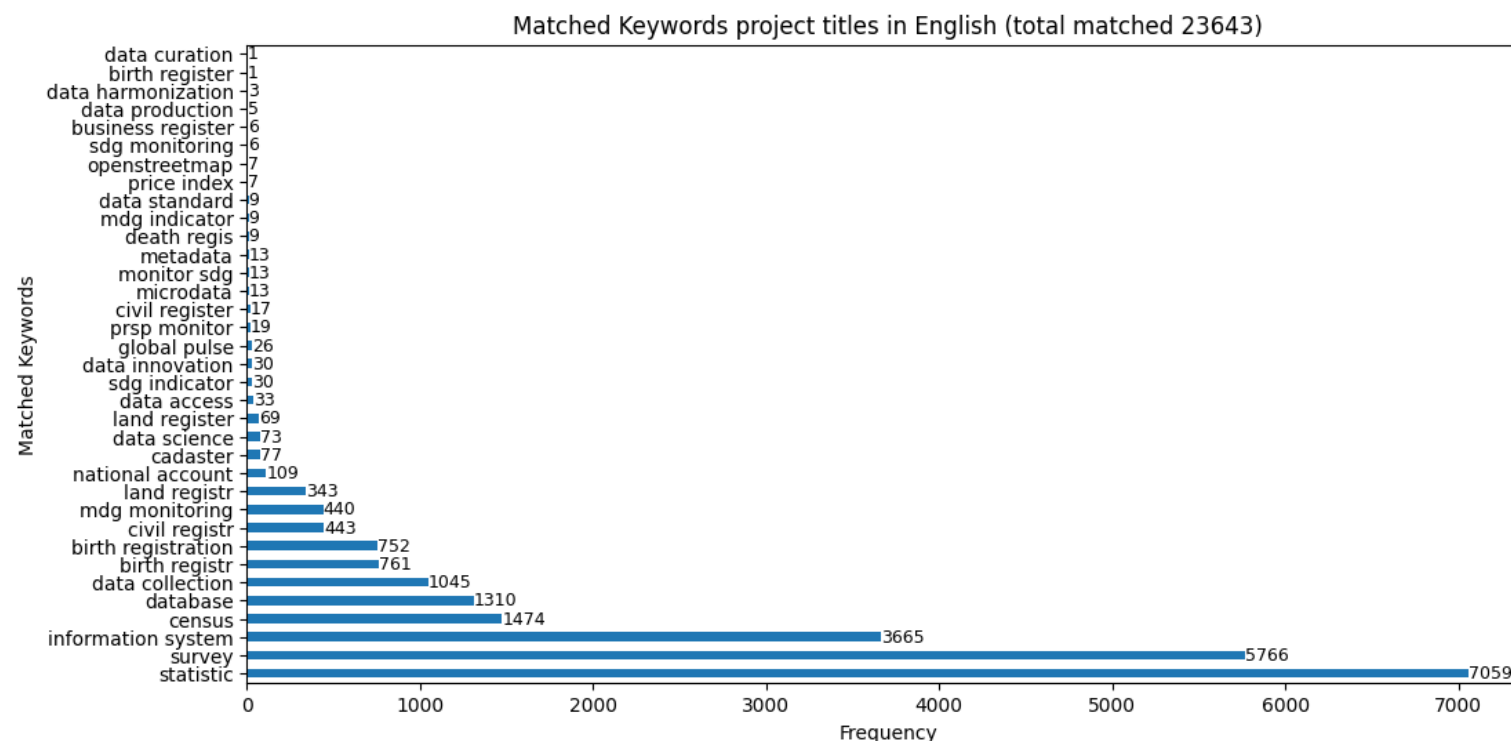
GBV
VAW
sdg5
WEE
SRHR
FGM

3. Current methodology

Project titles

A – Title pattern matching

number of matched keywords
with current keyword lists and
search in unique project titles for
entire CRS dataset (from 1973-
2023)



3. Current Methodology

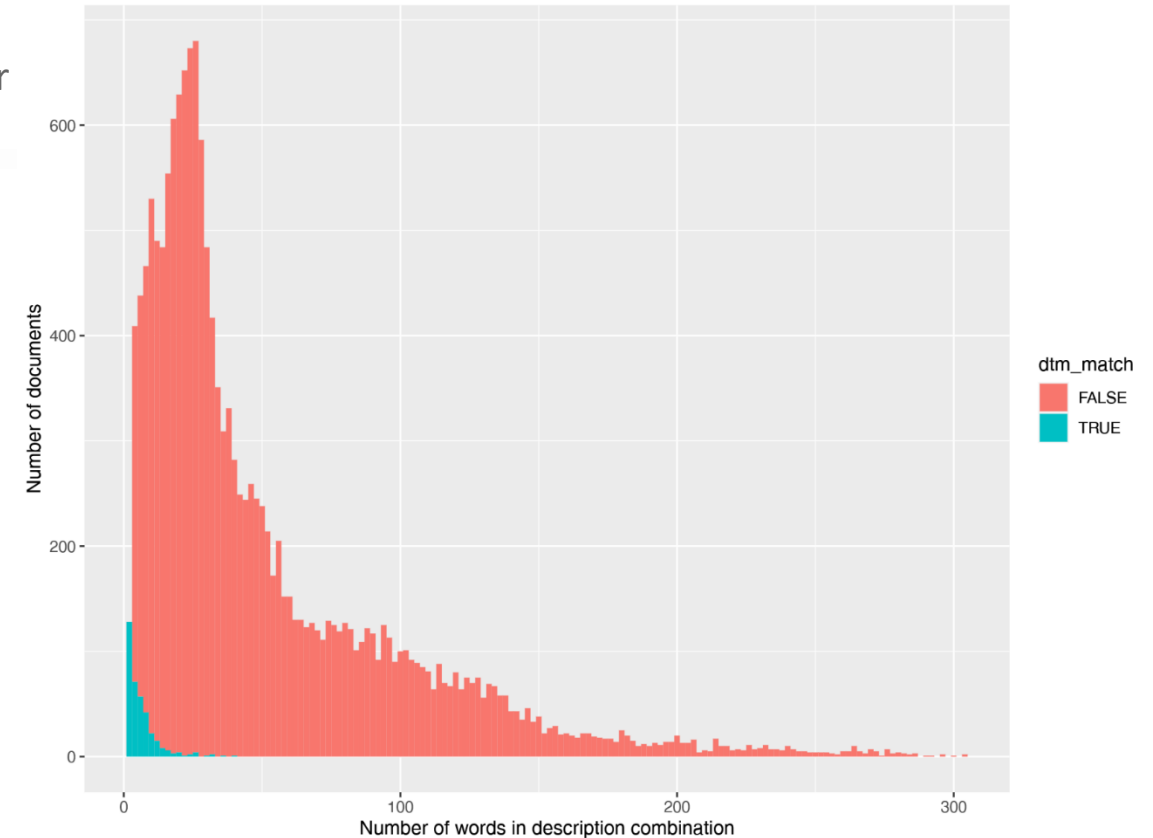
B – Text mining

Currently frequency-based approach using **odds ratio** of words that appear frequently in long description of statistics project

Issues with frequency-based approach:

1. **Independence of descriptions lengths and statistics match**
reasonable to assume that description length and statistics match are independent -> also long text can be statistical
2. **Accuracy much higher with modern approaches**
3. **For title matching: stemming instead of lemmatization**

plot from previous frequency-based approach, green projects with matched long description, orange not matched



4. Methodology Proposal

1. Classification Method

How to choose a model?

First, we need to sort our requirements according to their importance:

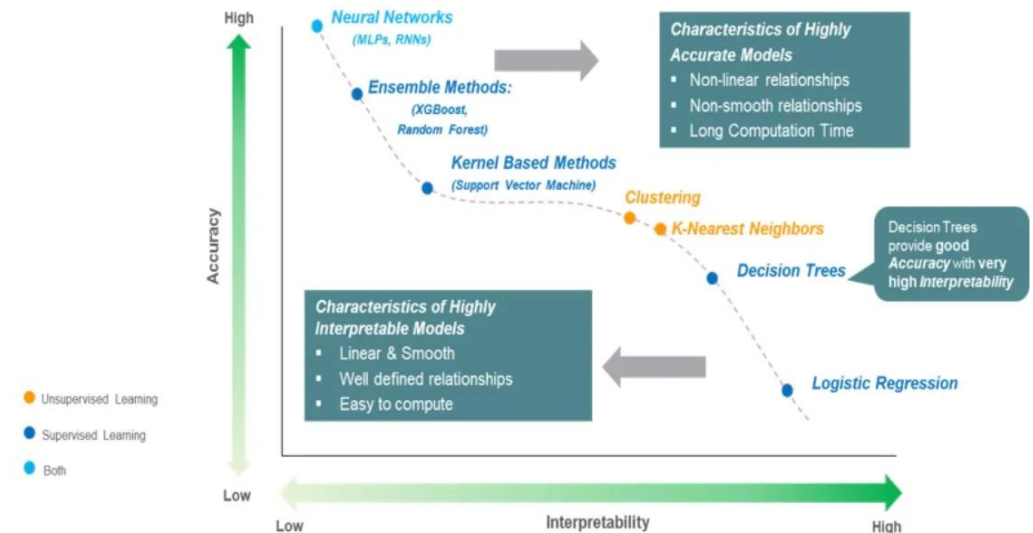
1. **Accuracy:** most important requirement – measuring funding accurately
2. **Multilingualism:** 22% of descriptions are not in English
3. **Interpretability:** insights into according to which words text is classified

Base requirements:

- **Text classification:** our task is a binary text classification, but this applies to all models
- **Positive-unlabeled learning problem:** this applies to all models and therefore doesn't need to be considered during model selection
- **Variance estimation:** it's possible to estimate the variance for all models

1. Most important requirement –Accuracy

in general, the more complex the model is, the more accurate it is. Since this is our most important criterion, **modern neural networks** provide the highest accuracy



4. Methodology Proposal

Which NN is suitable in our case?

Requirements:

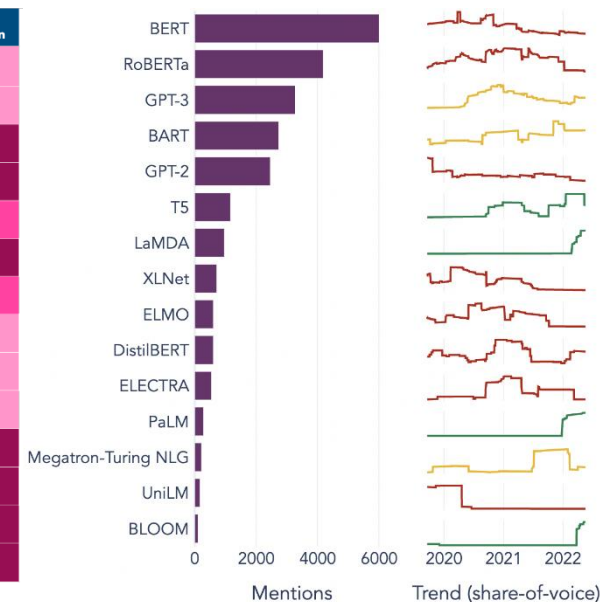
1. Accuracy
 2. Multilingualism
- Binary Text classification

To get an impression on which models are most frequently used in NLP and text classification, see their specifications and mentions on the right. By far, the open-sourced BERT (*Bidirectional Transformers for Language Understanding, 2019* - 128489 citations) by Google AI and all its adaptations (most prominently RoBERTa, DistilBERT, mBERT) are by far the most studied NNs for NLP in the literature.

Model	Core differentiator	Pre-training objective	Parameters	Access	Information Extraction	Text Classification	Conversational AI	Summarization	Machine Translation	Content generation
BERT	First transformer-based LLM	AE	370M	Source code						
RoBERTa	More robust training procedure	AE	354M	Source code						
GPT-3	Parameter size	AR	175B	API						
BART	Novel combination of pre-training objectives	AR and AE	147M	Source code						
GPT-2	Parameter size	AR	1.5B	Source code						
T5	Multi-task transfer learning	AR	11B	Source code						
LaMDA	Dialogue; safety and factual grounding	AR	137B	No access						
XLNet	Joint AE and AR	AE and AR	110M	Source code						
DistilBERT	Reduced model size via knowledge distillation	AE	82M	Source code						
ELECTRA	Computational efficiency	AE	335M	Source code						
PaLM	Training infrastructure	AR	540B	No access						
MT-NLG	Training infrastructure	AR and AE	530B	API						
UniLM	Optimised both for NLU and NLG	Seq2seq, AE and AR	340M	Source code						
BLOOM	Multilingual (46 languages)	AR	176B	Source code						

AR = Autoregression
AE = Autoencoding
Seq2seq = Sequence-to-sequence

Highly appropriate
Appropriate
Somewhat appropriate



most important modern NN in NLP ([Towards Data Science, 2022](#))

mentions of NNs online
([Towards Data Science, 2022](#))

4. Methodology Proposal

Which NN is suitable in a multilingual setting?

Requirements:

1. Accuracy
 2. Multilingualism
- Binary Text classification

From BERT, there are namely two adaptations that are tailor-made for multilingual data:

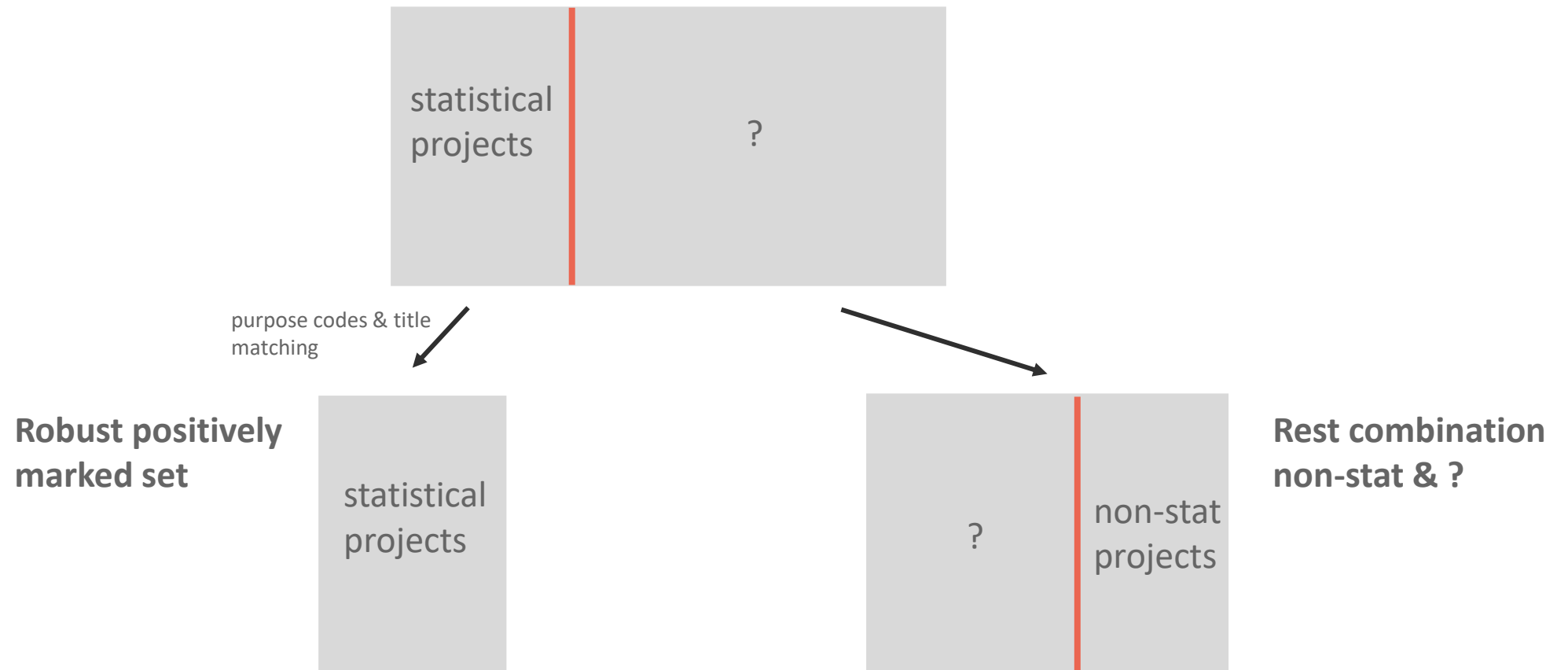
1. **multilingual BERT (mBert)**: trained on the entire Wikipedia of the 104 most utilized languages, it reaches almost the [accuracy as single language models](#)
2. **XLNet-RoBERTa (XLM-R)**: adaptation by Meta AI, in addition to Wikipedia training data, trained additionally on common crawl data of the internet and outperforms mBERT especially on low-resource languages

In our case, as it is a simple binary text classification task mostly in English with a large training set available, both will likely perform equally well. Either both can be tested, or the XLM-R can be chosen due to its performance on low-resource languages. For more information on their performance comparison, see [original XLM-R paper](#), as well as an [evaluation paper by Google](#).

Both are easily available through the Hugging Face API in the form of its *transformers* library, which is very easy to use and deploy.

4. Methodology Proposal

Positive-unlabeled learning problem



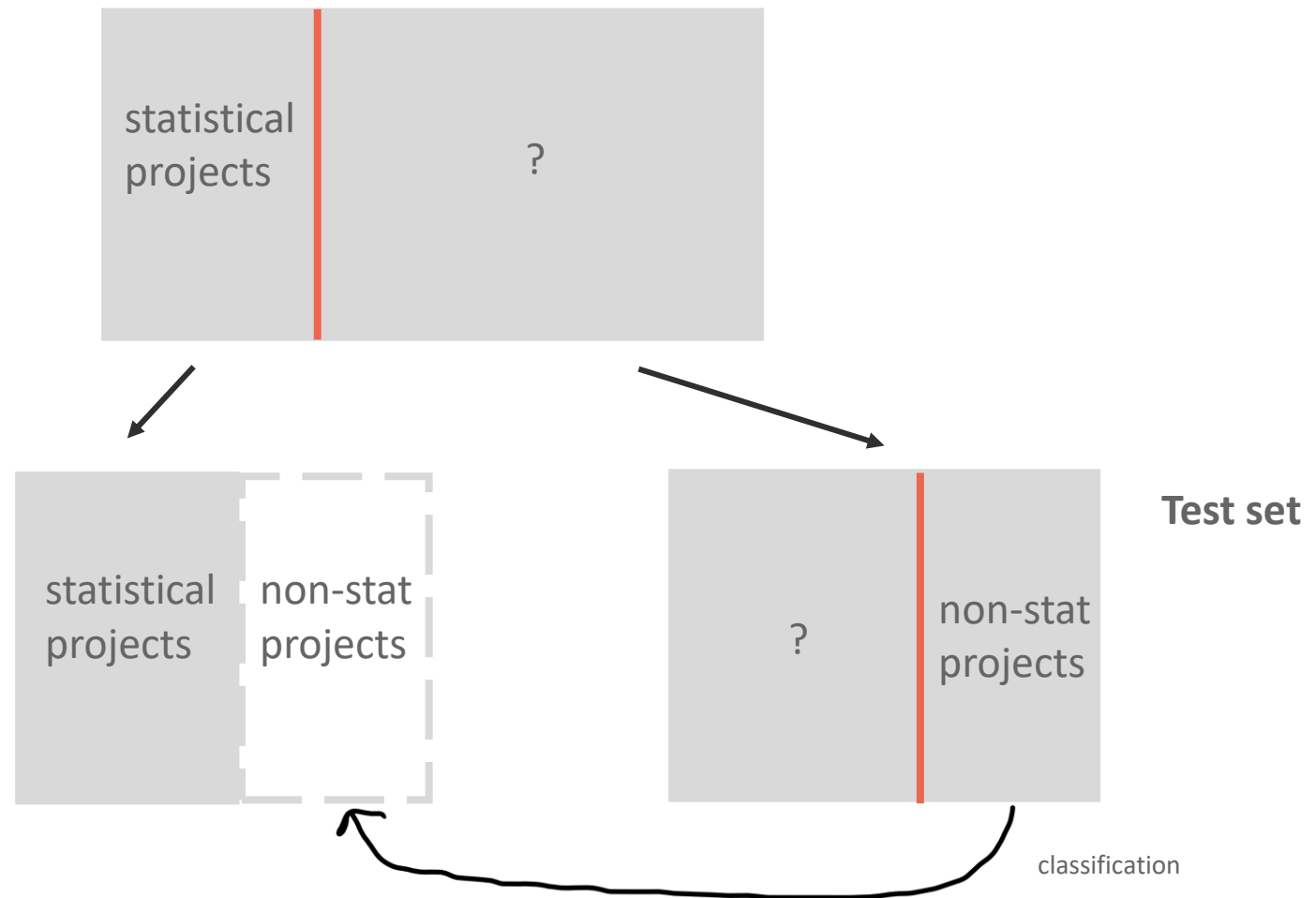
4. Methodology Proposal

Positive-unlabeled learning problem

solution:

1. Create training set with negatives as randomly sampled from ?
2. Train classifier based on training set
3. Recreate training set with classified negatives with high probability in class non-stat
4. Train classifier based on recreated training set

Robust training and validation set



4. Methodology Proposal

Interpretability

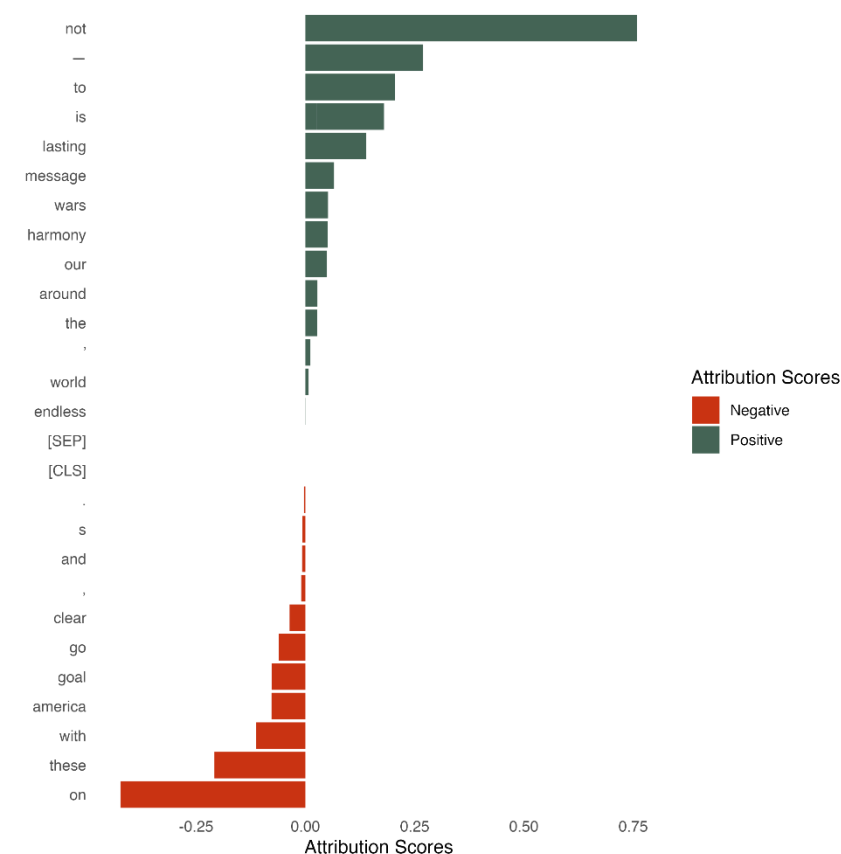
solution:

- **Word importance:** for every long description, a word importance is assigned to every word -> inspection of positively classified long descriptions
- **other methods:** SHAP values, attention visualization,...

word importance for multiple sentences (example for sentiment analysis)

Legend: ■ Negative □ Neutral ■ Positive				
True Label	Predicted Label	Attribution Label	Attribution Score	Word Importance
1	POSITIVE (1.00)	POSITIVE	0.82	[CLS] around the world our message is clear — america ' s goal is lasting harmony , and not to go on with these endless wars . [SEP]
Legend: ■ Negative □ Neutral ■ Positive				
True Label	Predicted Label	Attribution Label	Attribution Score	Word Importance
1	POSITIVE (0.97)	POSITIVE	2.12	[CLS] behind every one of our decisions are the voices and lives of the invisible masses whom we must defend , because we in turn were defended in the past . [SEP]
Legend: ■ Negative □ Neutral ■ Positive				
True Label	Predicted Label	Attribution Label	Attribution Score	Word Importance
0	NEGATIVE (0.89)	NEGATIVE	0.72	[CLS] that wealth , which right ##ly belongs to iran ' s people , also goes to shore up basha ##r al - ass ##ad ' s dictatorship , fuel yemen ' s civil war and undermine peace throughout the entire middle east . [SEP]
Legend: ■ Negative □ Neutral ■ Positive				
True Label	Predicted Label	Attribution Label	Attribution Score	Word Importance
0	NEGATIVE (0.98)	NEGATIVE	2.16	[CLS] everyone is tempted to follow their own law . [SEP]
Legend: ■ Negative □ Neutral ■ Positive				
True Label	Predicted Label	Attribution Label	Attribution Score	Word Importance
1	POSITIVE (1.00)	POSITIVE	1.63	[CLS] here too we will remain fully committed . [SEP]

word importance for one sentence



2. Summary new methodology

Preparation

1. **Bulk download from Website:** Automatic download of entire CRS dataset from OECD data explorer
2. **ID creation:** Create IDs for unique projects

A – Title pattern matching – **same** as before with **2 adaptations**:

For new methodology, we use the same title pattern matching procedure as before with 2 improvements (updated **language detection library**, **lemmatization** instead of stemming). The underlying assumption that a project is mainly focused on statistics if the respective keyword appears in the project title seems very reasonable. Therefore, by matching keywords in titles, we can introduce an additional focus on topics of our choice (e.g. data science,...) instead of just relying on the purpose codes. In addition, this would allow to adapt the methodology easily to new topics (e.g. AI, environment,...) by just exchanging the keyword lists (and potentially selecting different purpose codes).

B – Text mining with **new methodology**:

1. **Data augmentation:** easy to implement with readily available libraries. Measures could include: synonym swaps (to make training set more robust), translations from en to fr/es/de (more labeled data for minority languages)
2. **Create Training/validation set:** Use purpose code & title match as positive label, randomly sample negative labeled from rest
3. **Fine-tune NN:** download models and test if feasible on local machine/CPU. Likely necessary to use services which provide GPU compute but overall our task is not too resource intensive (XLM-R 550m parameters vs. 8B for smallest Llama which can be used for inference locally on CPU), therefore free services like Google colab could be sufficient (upgrade to 10\$ monthly plan available)
4. **Repeat 2.-4. to solve positive-labeled learning problem:** replace randomly sampled negatives in training set with samples with high predicted negative probability. *Alternatively:* Use a zero-shot classification (without fine-tuning) before creating the training/validation set -> fine-tune only once
5. **Choose threshold for inclusion:** every project will be given a probability 0-1 that it is a statistics project. A threshold could be chosen based on common metrics (accuracy, precision, recall, F1) or manual inspection of descriptions around for various thresholds.

4. Methodology Proposal

3. Comparison current vs proposed methodology

	Steps	Current methodology	Proposed methodology
Preparation		Manual Individual file download	Automatic bulk download from API
A – Title pattern matching	Clean text	Remove punctuation, numbers,...	Remove punctuation, numbers,...
	Language detection	CLD2	fasttext
	Keyword detection	Stemming	Lemmatization
	Create final markers	Use purpose code & title match	Use purpose code & title match
B – Text mining		Odds ratio for common words in matched long descriptions	Data augmentation
		Match acc. to threshold	Fine-tune NN
			Predict using fine-tuned NN

Implementation questions:

1. **Positive labeled learning problem:** Zero-shot classification vs fine-tuning twice
2. **Balance checks:** how to effectively test balance of training set?
3. **Technical NN training set-up:** any parameters that need special attention?